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Thermal conductivity and energy storage capacity enhancement and bottleneck of shape-stabilized phase change composites with graphene foam and carbon nanotubes

Zepei Yu [a], Daili Feng [a,b,*], Yanhui Feng [a,b,*], Xinxin Zhang [a]

[a] School of Energy & Environmental Engineering, University of Science and Technology Beijing, Beijing 100083, China

[b] Beijing Key Laboratory of Energy Saving and Emission Reduction for Metallurgical Industry, University of Science and Technology, Beijing, Beijing 100083, China

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ABSTRACT

A systematic, carbon-based composite phase change materials with substantial increase of the thermal conductivity and energy storage density was assembled by encapsulating PEG into graphene foams (GF), CNTs and hierarchical porous materials derived from GF and CNT. The influence of preparation approaches on the microstructures, thermal conductivity and thermal energy densities of composites were assessed by various characterization tools. The results obtained indicated that graphene foam-CNT hybrid structures (GCNT) composites exhibit excellent thermal conductivity, 118% higher than pure PEG and 89.5% higher relative to that of GF/PEG@CNT (GPC) composites. The content and interconnected hierarchical porous carbons enhance thermal transfer in composites during the process of phase change. Notably, GCNT@PEG (GCP) had a maximum of up to 80 wt% PCM loading. This composite has a melting latent heat of 128.7 kJ/kg, up to 39.3% and 87.5% higher than those of other composites. CNTs have significantly enhanced the thermal transport of composite PCMs, but the interfacial thermal resistance of CNTs and graphene foam is the bottleneck of heat transfer of composite PCMs. The current results create the conditions for the application of novel hierarchical porous carbon composite PCMs for high-energy density, thermal energy conversion, and shape-stabilized PCMs (ssPCMs) with improved thermal conductivity.

1. Introduction

The imminent reduction in fossil resources will induce shortages of both energy and carbon sources. To overcome future energy crises, thermal energy storage (TES) is anticipated to play a vital role in the energy production of the future, especially for renewable energy systems in which the discontinuities and unpredictable nature of energy sources are major concerns [1–6]. To improve the thermal energy storage efficiency of TES systems, energy storage methods such as PCMs have attracted the greatest attention due to their potential of storing and releasing a significant amount of latent heat over the process of phase change [7–12].

Among PCMs, organic PCMs have attracted remarkable attention over inorganic PCMs thanks to their excellent chemical stability, adjustable phase transition temperature, low supercooling, high storage potential of latent heat, and general lack of toxicity [12–14]. Polyethylene glycols (PEG) have been used for a wide variety of applications given their high latent heat, suitable phase transition temperature, self-nucleating behavior, low vapor pressure, and excellent physical and chemical stability [15–16]. In the thermal management industry, they have been widely used to maintain a constant required temperature for temperature-sensitive products during transfer [17–18]. However, their applications in TES are greatly restricted due to poor thermal conductivity, leakage, and flammability [19]. In order to overcome issues of leakage and low storage capacity to develop high-performance phase change materials, several studies have led to the development of a sequence of materials called shape-stabilized PCMs. Tang et al. [20] effectively prepared PA-CA/diatomite shell composites with an energy storage capability of 98.3 kJ/kg. Similarly, Alva et al. [21] introduced silica as a supporting scaffold for MA–PA eutectic mixtures for thermal energy storage composite PCMs and demonstrated a high storage capacity. However, the utilization of ssPCMs for energy storage still needs further improvement to overcome another major bottleneck of heat transfer enhancement: low thermal conductivity. This low thermal

* Corresponding authors.
E-mail addresses: dlfeng@ustb.edu.cn (D. Feng), yhfeng@me.ustb.edu.cn (Y. Feng).

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conductivity of PCMs (generally lower than 1 $W\cdot m^{-1} K^{-1}$) leads to poor adsorbing and releasing rates for the thermal energy storage system, in turn resulting in heat charging and discharging times that are unacceptable for practical applications [22]. To solve this problem, many researches have aimed to improve the thermal conductivity of the phase-change composites by using high thermal conductive fillers. A number of heat transfer enhancement methods have also been studied previously [23]. Two kinds of methods are typically used: mixing PCM with solid particles [24] or dwelling PCM into porous matrix materials with excellent heat transfer performance [25].

Many high-thermal-conductivity particles, including carbon black nanoparticles [26], silicon dioxide [27], carbon fibers [28], carbon nanotubes [29] and Al_2O_3-loaded expanded vermiculite [30] have been used to fabricate PCM composites according to the former method. More recently, carbon allotrope materials, for instance CNTs (carbon nanotubes), as well as CNFs (carbon nanofibers), have attracted remarkable attention, as their high aspect ratios significantly strengthen thermal conductivity as forecasted by effective medium theory [31–32]. Soy wax PCMs were mixed with CNFs at different mass ratios and the composites exhibited excellent thermal conductivity relative to standalone PCMs [33]. However, the thermal conductivity values of these researches are not sufficient for practical applications. With respect to the latter method, many porous materials with great thermal conductivity have been employed, including porous carbon materials like carbon nanotube sponges [19], graphite foams (GF) [34–35], and graphene aerogels [36]; metal foams including copper [37], Iron-Nickel [37], Ni [38], and Al [39]. Out of these porous materials, carbonaceous materials with porous structures have been noted for their excellent heat transfer performance. A 3D graphene material was applied to enhance the heat transfer of paraffin and to enhance the shape stabilization [40]. Huang et al. [41] introduced expanded graphite into palmitic acid–stearic acid to enhance the thermal conductivity by up to 5.6 times compared to pure PCM. For the impregnation PCM into porous material method, even when the porous materials have an excellent interconnected structure for thermal transport, the thermal conductivity of the matrix materials limit the thermal conductivity of the composite PCMs.

In this study, in light of these two methods, we combine the porous structure with nanoparticles to enhance the thermal conductivity of composite PCMs. It can be concluded from aforementioned researches that carbon materials are the most suitable material for enhancing the thermal conductivity of PCMs because of their initial high thermal conductivity. The incorporation of CNTs and polyethylene glycol (PEG) into systematically prepared graphene foams (GF) for enhanced heat transfer and storage performance was also investigated. The choice of PEG stemmed from its high fusion enthalpy, physical, as well as chemical stability, and suitable phase transition temperature. To mitigate PEG leakage and enhance thermal conductivity, CNTs and graphene foams were employed as thermally conductive fillers and matrix materials, respectively, in different mass ratios, and their effects on thermal properties were analyzed. The use of composites could adjust high ambient heat flow and exhibits great potential in applications of thermal protection and management. As the forecast of interfacial thermal resistance acts an important role in the heat transport of composite PCMs, this paper also performed molecular dynamics (MD) simulation using nonequilibrium method to calculate the interfacial thermal resistance of all samples. The enhancement and bottleneck of thermal transport of composite PCMs were investigated to gain insight into shape-stabilized phase change materials (ssPCMs) designing and TES manufacture by utilizing a combination of conductive fillers and porous materials.

2. Experimental methods

2.1. Materials

Carbon nanotubes (CNTs, diameter 10–20 nm and length 0.5–2 μm) and graphene foam (specific surface area: 267.5 m^2 g^{-1}) were commercially provided by Xianfeng Nanotech Company (Nanjing, China). Polyethylene glycol 2000 (HO–(CH2CH2O)n-H), purchased from Aladdin Reagent Company (Shanghai, China), was chosen as the PCM. All the chemicals were pure analytically and utilized without additional purification.

2.2. Synthesis of composite materials

GCP composites were prepared through the solution impregnation approach according to our previous study. This approach has proven useful for the fabrication of PCM composites [16,42]. The preparation process for composite PCMs is illustrated in Fig. 1.

The GPC composites were assembled as indicated: First, 7.5 mg CNTs were totally dispersed in 200 mL alcohol under ultrasonic vibration for 5 h (300 W). Next, CNTs solution and 35 mg PEG 2000 were added to a quartz cup. The mixture was stirred for 40 min at 75 ℃. At this point, the PEG was totally melted in the alcohol, and the CNTs were totally dispersed in mixture. An oven was used to dry the sample at 100 °C for 12 h. Samples obtained were called "PC." Next, 42.5 mg PC, 7.5 mg GF and 200 mL alcohol were added to a quartz cup. The mixture was stirred for 40 min at 70 ℃ to obtain GPC solution. Likewise, the mixture was dried in a thermostatic drying cabinet at 100 °C for 12 h. The final products were GPC ssPCMs with PEG loadings of 70 wt%. In the same way, GF@PEG (GP) composites were prepared containing 70 wt% PEG.

The maximum loading of GCP was 80 wt% obtained in previous study [43]. To test the maximum loading of the GPC and GP, the 80 wt%, 70 wt%, 60 wt%, 40 wt%, 25 wt% and pure PEG samples were placed on filter paper and heated in a drying cabinet at 100 °C for 2 h. Fig. 2. shows the images of the samples at ambient temperature and 100 °C.

In case of grease on the filter paper, PCMs had melted to leak from the matrix materials. Herein, the maximum loadings of GPC and GP were both 70 wt%.

2.3. Characterization

A scanning electron microscope (SEM, JEOL, JSM-7001FA) was employed to observe the microstructures of GCP, GPC and GF-based composite PCMs. Nitrogen adsorption tests were performed to measure surface area and pore volume taking measurements with a porosity analyzer (Micromeritics ASAP 2460, 77.3 K). Crystalline products were characterized by carrying out XRD diffraction analyses. The laser flash analysis (LFA) method is a robust way to determine the thermal conductivity of solid materials [44–45]. Thermal conductivity was measured by the LFA method with LFA447 (Netzsch GmbH, Selb, Germany). The thermal conductivity of GPC can be calculated as

$$k_{GPC} = \alpha_{GPC} \cdot C_{P,GPC} \cdot \rho_{GPC} \tag{1}$$

Each samples was shaped at a pressure of 5 tons into a disc with d = 12.5 mm and h = 2–3 mm for LFA method to measure thermal diffusivity. All composite materials were determined at room temperature. Differential scanning calorimetry (DSC, TA SDT-Q600) was used to characterize latent heat and phase transition temperature of composite PCMs at a heating, as well as cooling rate of 10 °C/min [46].

3. Simulation methodology

3.1. Model

The models of all samples were constructed in Materials Studio. In this study, (6, 6) single-walled CNTs with length 42.5 Å were used for calculation simplification. The amorphous carbon model was constructed refers to the research of Feng et al. [47] As shown in Fig. 2, two 64.1 Å×63.5 Å×10 Å parallel layers of amorphous carbon represent the graphene foam matrix. The four CNTs were placed along the z-axis or y-axis in GCP and GPC composite models, respectively. As a comparative

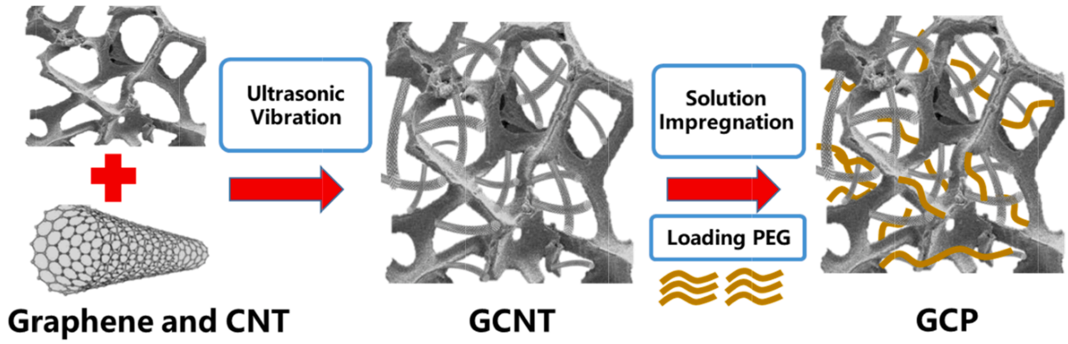


Fig. 1. Schematic diagram of the process of preparing composite phase change materials.

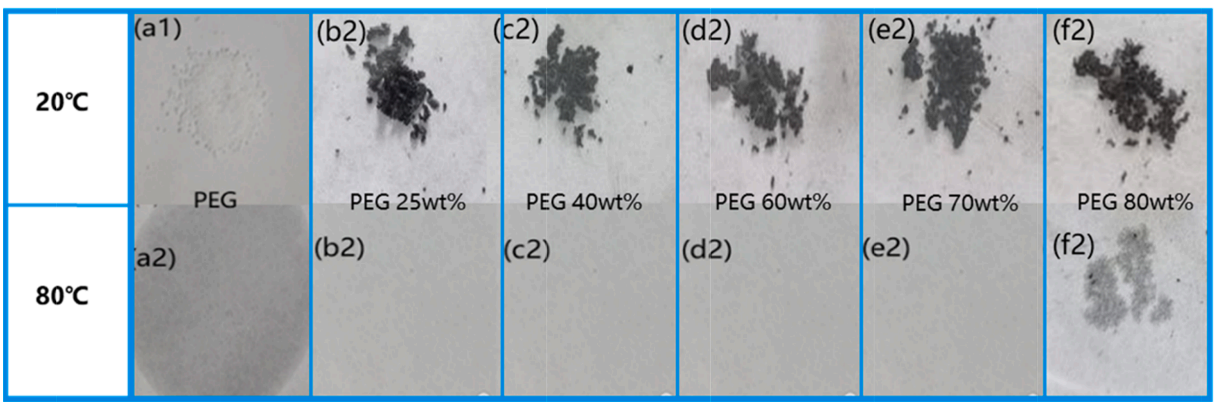


Fig. 2. Schematic diagram of the process of preparing composite phase change materials.

numerical example, there was no CNTs in GP composite model. For model limitation and computational simplicity, the number of PEG lines was limited to 32 and the weight loading of three models was 25 wt%.

3.2. Force field

In MD simulations, the PCFF force field was employed to describe the interanctions between PEG core atoms, and the Tersoff potential [48] was used for the interactions between C-C atoms in GF and CNTs. The LJ potential was assigned to model non-bonding interactions between matrixes and PCMs, and the LJ parameters were computed basing on Lorentz-Berthelot mixing rules. The parameters of C atoms in GF and CNT were ε=0.054 kb/K, R_{min} = 4.01 Å, and the parameters of atoms of PEG molecules referred to the paper [47].

4. Results and discussion

4.1. Morphology and microstructure

Fig. 3 shows the morphology and microstructure of the samples as visualized using scanning electron microscopy (SEM). A well-interconnected GCNT porous network structure with high porosity can be clearly observed in Fig. 3(a). Additionally, the GCNT matrixes exhibited a relatively smaller sized particle, creating a favorable working climate for transport of heat speed in samples during the process of phase transition. Meanwhile, the GPC composite appeared to be in a sheet accumulation state. The pores in the carbon matrixes is totally or partially filled by PEG, as shown in Fig. 3a-c. Noticeably, it was observed that the surface of three samples became compact at maximum loading.

The porosity characteristics of the matrix materials were assessed

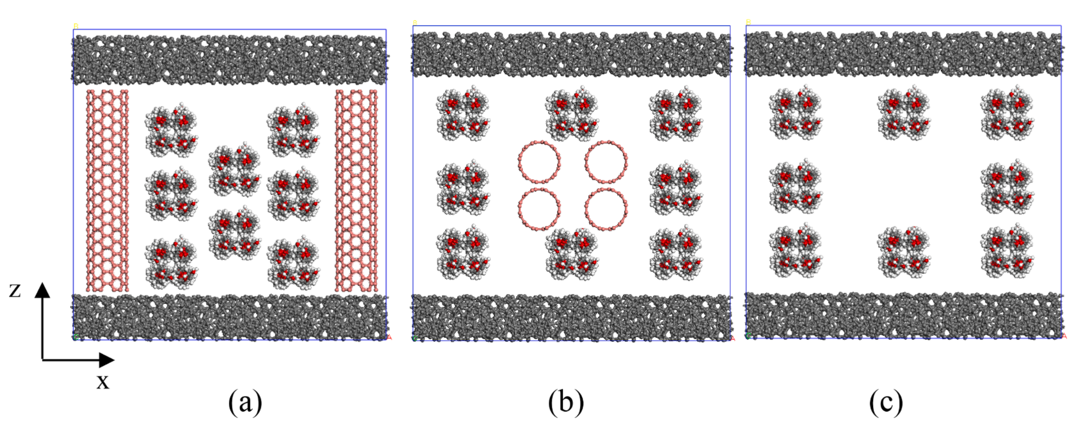


Fig. 3. Models of GCP、GPC and GP (25 wt%).

using N_2 sorption analysis (Fig. 4). It has been previously reported that GF and GCNT show similar type IV isotherms, with dramatical increases at high pressure indicating that the GF and GCNT samples have micro-, meso-, and macropores. Additionally, the sharp escalation of the isotherm in the low pressure region disappears in the isotherm of CNT, revealing that there are no micropore characteristics for CNT [43].

Table 1 exhibits a summary of the Specific surface area (BET method), overall pore volume (BJH method) and average pore diameter. GF had a high specific surface area (2067.01 m^2/g) and a good pore volume (0.92 cm^3/g), with an average pore diameter centered at 19.2 nm. GCNT exhibited a lower pore volume (229.63 cm^3), but the highest loading rate of GCNT; the rate reached 80 wt%, suggesting a favorable climate for notable PCM loading. Almost no micropores or mesopores were observed after incorporating PEG into the composites Table 2.

4.2. Morphological and structural characteristics of composite PCMs

XRD patterns of GF, GCNT CNT and their composites were assessed to investigate the influence of matrix materials on PEG crystallization. As shown in Fig. 5, the amorphous nature of carbon in GF and GCNT is indicated by a peak at 21.3°. The XRD patterns of CNTs feature a peak at 26.51° which can be attributed to the hexagonal graphite structure, which suggests the lattice planes of (0 0 2). There are only two peaks in GCNT, but no shifts of peaks or other new peaks were observed, suggesting that excellent physical compatibility and no chemical reaction or interfacial bonding occurs.

Typical diffraction peaks of pristine PEG 2000 are detected at 19.2°and 23.4° [15,49]. Three composite systems feature peaks corresponding to both PCMs and matrixes. There are no shifts of typical peaks indicates that crystal structure of PEG is maintained even after impregnated into matrix materials [50–51].

4.3. Thermal conductivity improvement

Thermal conductivity is a key parameter in characterizing the efficiency of ssPCMs at transferring thermal energy away from the source, influencing the rate of energy absorption and excretion [7,52]. Using GF and CNT as the matrix materials is expected to strength the thermal transfer of the composites. This is exceedingly beneficial for ssPCMs applications, as it reduces the local overheating and energy storage and release time. To evaluate the heat transfer reinforcement of the composites as compared to pristine PEG, the LFA method was performed following the progress mentioned in Section 2. Fig. 6a shows the thermal conductivity of pure PEG and three composite systems with corresponding carbon matrixes. After the introduction of PEG into GCNT and GF/CNT, thermal conductivity was enhanced to 0.351 and 0.665 W/mK, respectively, with corresponding increases of 15.1% and 118% over pristine PCM, allowing an improved heat transport rate for the prepared samples. Additionally, thermal conductivity decreased to 0.256 W/mK when GF was introduced, indicating that some macropores were still present in the GF matrix. These macropores were unable to adsorb and immobilize PCMs. Meanwhile, for GCNT, the CNT was introduced into GF to produce more micropores and mesopores, generating more heat transfer channels. Fig. 6b shows that, as loading rate is increased, the thermal conductivity of GCP composites increases gradually. It is noted that GCNT shows relatively excellent thermal conductivity performance in composite PCMs.

Table 1
Structural parameters of samples.

Samples	BET surface area (m^2/g)	BJH pore volume (cm^3/g)	Pore diameter (nm)
Graphene foam [43]	2067.01	0.92	19.2
GCNT [43]	229.63	0.76	19.2 15.3
80 wt% GCP [43]	4.92	0.004	–
70 wt% GPC	5.28	0.005	–
70 wt% GF	5.21	0.004	

Table 2
Thermal characteristics of pure PEG and three composite materials.

Sample	T_m (°C)	ΔH_m (J/g)	T_f (°C)	ΔH_f (J/g)	Supercooling (°C)
Pure PEG	51.11	175.6	33.50	173.2	17.61
80 wt%GCP [43]	50.31	128.7	37.59	126.52	12.72
70 wt%GPC	48.25	68.64	33.63	60.84	14.62
70 wt%GP	48.16	92.40	33.94	88.92	14.22

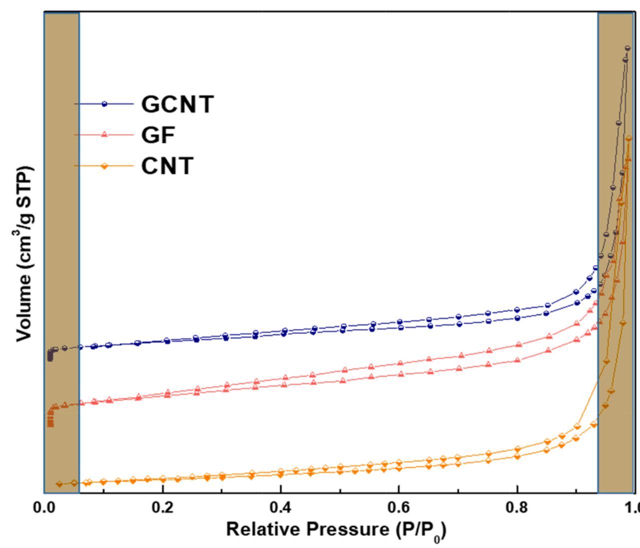
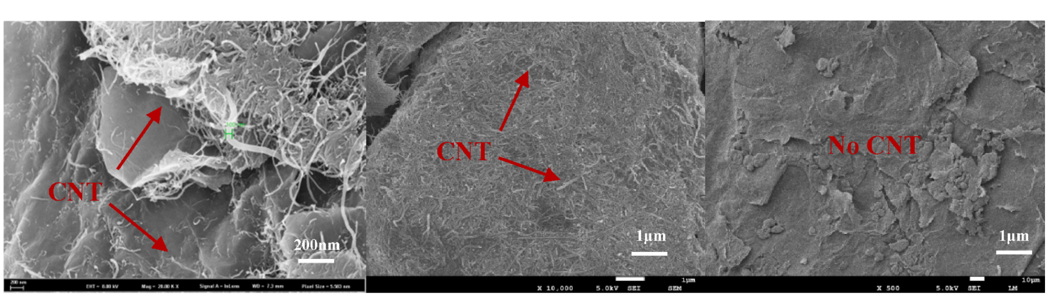


Fig. 5. N_2 adsorption–desorption isotherms of graphene foam, GCNT [43], and CNT.



(a) (b) (c)

Fig. 4. SEM images at different magnifications of (a) GCP, (b) GPC and (c) GP.

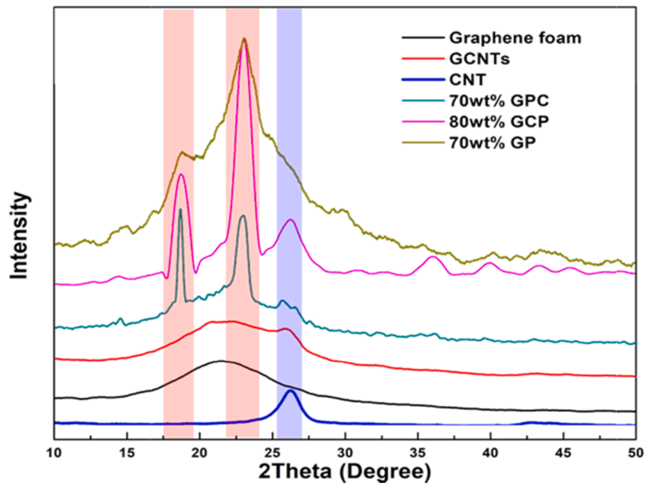


Fig. 6. Powder XRD measurements was employed to characterize the crystal structures of the samples.

4.4. Shape-stabilized characteristics of composite PCMs

4.4.1. Thermal properties of composites

Thermal characteristics, such as phase transition temperature along with latent heat of the pure PEG and composite PCMs, were determined using DSC. The values of heating and cooling curves for pure PEG were compared with those of composite materials and table 2 shows a summary of the thermal properties of pure PEG and composites. Fig. 7 exhibits the thermal characteristics of pristine PEG and composite PCMs. During the process of heating and cooling, all samples showed one solid–liquid and liquid–solid transition peak, respectively. Given carbon materials don't show phenomenon of phase change, it is the pure PEG that undergoes phase transition and contributes to the latent heat. The melting temperatures of the composite PCMs were consistent with that of pristine PCM, but were slightly lower, possibly attributable to the effects of physical interaction between the PEG and internal pore wall of the matrix materials. The results agree well with chemical compatibility and structural analysis mentioned above.

4.4.2. Supercooling suppression

Supercooling is prevalent during the process of freezing and serves as a major condition for triggering crystallization. However, since it can result in a decrease in freezing temperature and retard the freezing of PCMs, the serious supercooling is a non-negligible issue, indicating that the thermal energy cannot be discharged in time. The mismatch of phase transition temperature between charging and discharging reduces the efficiency of TES, thus the potential application of ssPCMs is seriously limited. After assembling, the melting temperature of the composites is slightly decreased attributed to the scale effect [54].

Interestingly, as shown in Fig. 8, the supercooling rate of pure PEG is 17.61 °C, and this value narrows with the introduction of matrix materials. The difference between the melting and cooling temperatures of PEG was significantly reduced under the influence of GCNT, with a suppression ratio of nearly 28%, compared with GF and GF/CNT.

4.4.3. Heat storage capacity and efficiency

Generally speaking, the introduction of a porous skeleton has a negative effect on the crystallinity of PCM and results in reduced heat storage capacity. Here, the crystallization efficiency θ was used to reflect the impact of GCNT on crystallization of PCM confined in the pores:

$$\theta = \frac{H_{m,comp}}{wt\% \times H_{m,PCM}} \times 100\% \tag{2}$$

where $H_{m,comp}$represents latent heat of ssPCMs acquired by the DSC test, wt% designates the mass percentage of PCM in composite, and $H_{m,PCM}$refers to the latent heat of the pure PCM. Crystallization efficiencies derived from melting enthalpies and cooling enthalpies are presented in Fig. 8a. Furthermore, the encapsulation fraction R and heat storage efficiency E were evaluated by Eqs. (3) and (4):

$$R = \frac{\Delta H_{m,comp}}{\Delta H_{f,PCM}} \times 100\% \tag{3}$$

$$E = \frac{\Delta H_{m,comp} + \Delta H_{f,comp}}{wt\% \times (\Delta H_{m,PCM} + \Delta H_{f,PCM})} \times 100\% \tag{4}$$

where$\Delta H_{m,comp}$ refers to the melting latent heat and$\Delta H_{f,comp}$ represents the crystallization latent heat of composites;$\Delta H_{m,PCM}$ and$\Delta H_{f,PCM}$respectively signify the melting and cooling enthalpies of PEG. As shown in Fig. 9, for 80 wt% GCP, 91.6% of confined PEG was crystallized, with an encapsulation fraction of 73.29%. The corresponding heat storage efficiency reached 91.46%, much higher than those of the other two composites. Due to the high adsorption capacity of GCNT, more PCMs were stored in the pores. This strengthened intermolecular interactions within the PCM and weakened interactions with GCNT, thereby leading to high crystallinity. Furthermore, thermal energy could be efficiently stored and released over a melting-freezing cycle due to the high heat storage efficiency. These results confirm

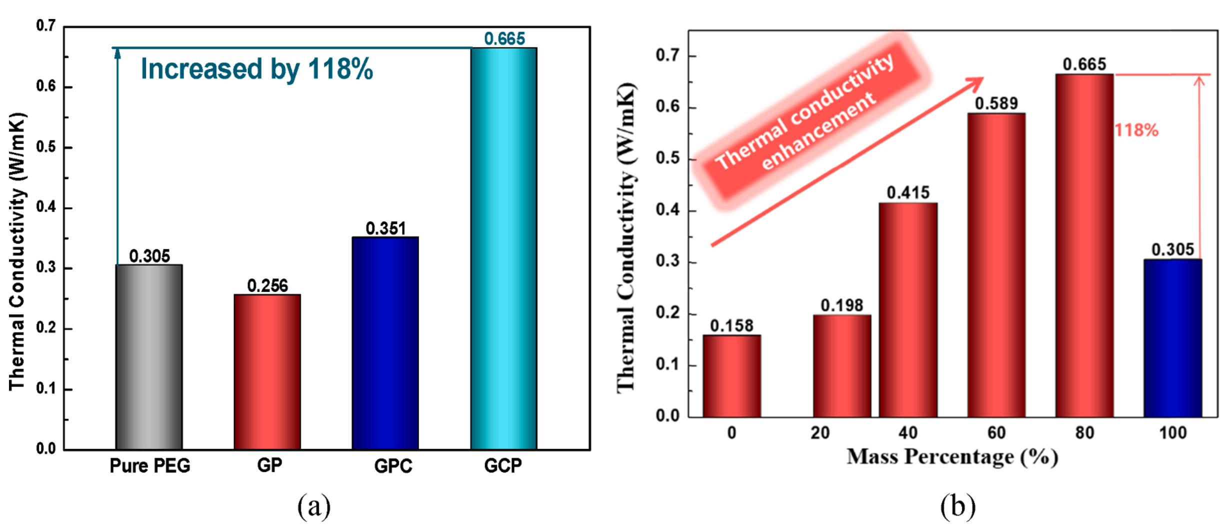


Fig. 7. Thermal conductivity of (a) pure PEG and three composite PCMs and (b) GCP with different loading rates.

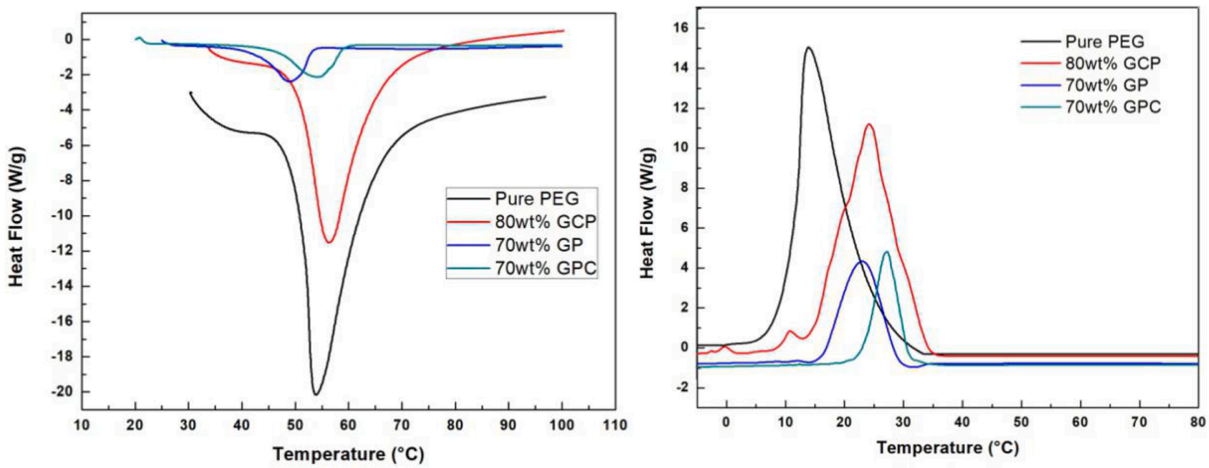


Fig. 8. Thermal conductivity of (a) pure PEG and three composite PCMs and (b) GCP with different loading rates.

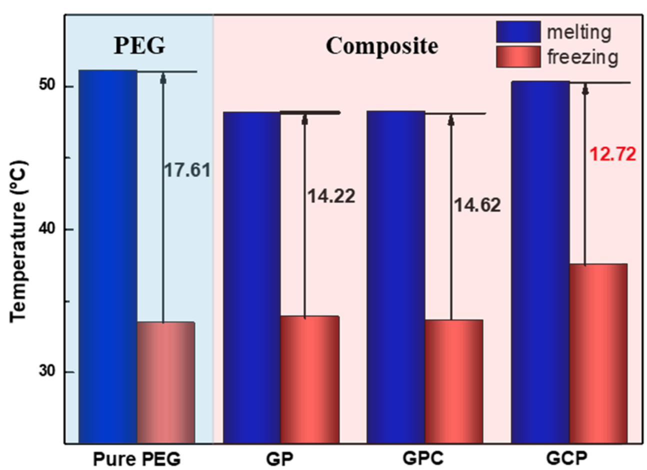


Fig. 9. Supercooling rate of pure PEG and three composites.

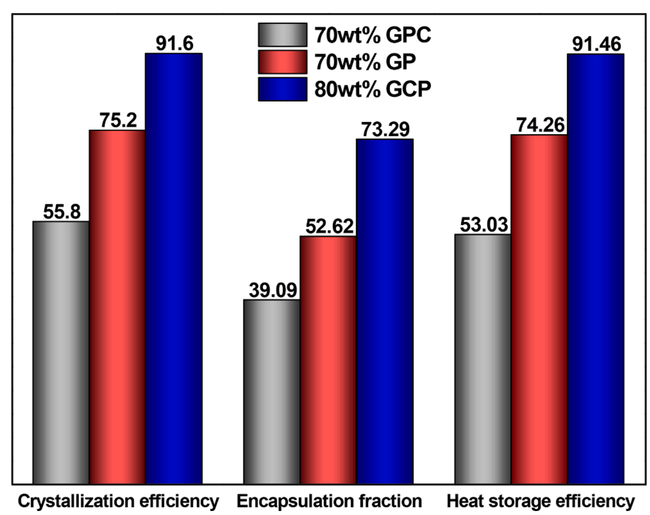


Fig. 10. Heat storage performance of three composites.

that, although a fraction of PCM still failed to crystallize, GCNT minimized negative effects on crystallization that are attributed to the presence of macropores, thus guaranteeing high heat storage performance.

With these results, the composite materials are competitive with recently reported composite materials [55–56]. Among the most commonly used PCMs, PEG has been packed with many nanoporous materials, corresponding storage performances are displayed in Fig. 10 and compared with those of GCNT composites. GCNT shows obvious advantages over other nanoporous materials due to its hierarchical pore structure, as anticipated. The improvement of thermal conductivity along with latent heat can most likely be due to the interconnect structure of GCNT with appreciable surface area that provide leading heat transfer pathways, and intermolecular interactions between GCNT and PEG.

4.5. Interfacial thermal resistance

According to the conclusion in section 4.4, the adding of CNTs can significantly enhance the heat transfer of composite PCMs, but the influence of interfacial thermal resistance cannot be ignored. Through the calculation of interfacial thermal resistance, we further explore the effect of the adding of CNTs on the thermal transport of composite PCMs. The thermal resistance is described as:

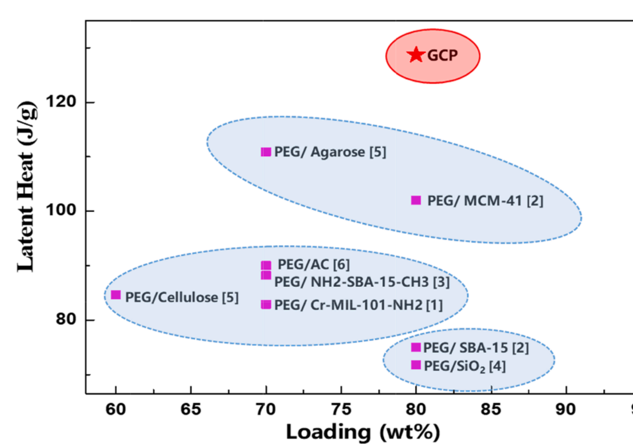


Fig. 11. The thermal energy storage capacity of PEG in GCNT compared with other nanoporous skeletons.

$$R = A\frac{\nabla T}{q} \quad (1)$$

where R is thermal resistance, A is the sectional area, ∇T is temperature gradient and q is heat flux.

The temperature distribution of two systems before and after adding CNT at 300 K ambient temperature is shown in Fig. 11. We can see that there is an obvious temperature step in the hollow region of the composites, indicating that there is notable interfacial thermal resistance in this area. The temperature data is further processed and the results were shown in Fig. 12 (a). The interfacial thermal resistance of GCP and GP are 1.7×10^{-8} m^2 K/W and 5.2×10^{-8} m^2 K/W, respectively. From the figure, we can see that the interfacial thermal resistance of GCP is reduced by 67.3% compared with that that of GP, implying that the adding of CNTs constructs the heat transfer bridge between the PCMs and the matrix materials, and the heat transfer enhancement effect is remarkable.

In addition, the effect of CNTs in different places on the heat transport of composite systems is investigated. The experiment results show that GCNT assembly method improves the heat transport of composite PCMs significantly higher than PCNT. In this section, the influence of different assembly methods of CNTs on the heat transfer of composites is explored from the micro level by calculating the interfacial thermal resistance of different systems. The calculation results are shown in Fig. 12 (b). The interfacial thermal resistance of GCP system is reduced by 66.8% compared with that of GPC. The assembling method of CNTs and graphene foam is obviously better than that of CNTs and PEG. This is because the CNTs and GF can build three-dimensional skeleton network and form effective heat transport channels. CNTs are coated in PEG molecules, which cannot be connected with each other to form heat transport network. Moreover, CNTs are easy to agglomerate in the process of phase change, producing a large number of new interfaces, resulting in limited heat transfer enhancement of composites, which is unable to achieve the expected enhancement.

Additionally, as shown in Fig. 13, we compare the thermal resistance of GPC and GP. It is found that the assembly of CNTs and PEG molecules reduces the interfacial thermal resistance of composite by only 2%, indicating that the dispersion and assembly of CNTs in PEG has finite improvement on thermal conductivity of composite systems.

In conclusion, CNTs have significantly enhanced the thermal transport of composite PCMs, but the thermal conductivity of composites with CNTs is still not ideal from the perspective of experiment or simulation. Therefore, the interfacial thermal resistance of GCNT, pillar graphene (PG) [7] and their composite PCMs were compared further. As shown in Fig. 14, the thermal resistance of the GCNT is 30 times that of PG. After assemble with PEG, the thermal resistance of GCP is larger than that of PG by an order of magnitude. In other words, the addition of CNTs reduces the interfacial thermal resistance of the composite PCMs, but the CNTs and the graphene foam are assembled on the basis of the intermolecular interaction force, while in the PG, CNTs and multilayer graphene are connected by chemical bonds, and their interaction force is far larger than the intermolecular interaction force. From the previous studies [15,57–59], it is known that the interaction force between PCMs and matrixes has distinct roles in heat transport. Therefore, the thermal resistance of the CNTs and graphene foam limits the further reinforcement of the thermal transport capacity of the composite PCMs. Functional group modification methods or directly growing CNTs on the surface of graphene foam can be employed to enhance the molecular interaction between CNTs and graphene foams, thus improving the thermal conductivity of composite PCMs [60].

5. Conclusion

In this study, carbon-based shape-stabilized composite PCMs were assembled using GF, CNT and GCNT as encapsulating agents. The phase transition latent heat of pristine PEG was measured (175.6 J/g) with a melting temperature of 51.11 °C. Composite materials were synthesized by varying mass ratio of pure PEG and the matrix material. Large loading percentages of up to 80 wt%, 70 wt% and 70 wt% were realized for GCP, GP and GPC, respectively. Meanwhile, all assembled composite PCMs showed good chemical compatibility, as confirmed by XRD results. For the assembled composites, GCP showed a notable thermal energy storage capacity of 128.7 k/g with a melting temperature of 50.31 °C. The prepared composite PCMs exhibited remarkable thermal conductivity up to 118% larger than pure PEG and about 89.5% higher than that of GPC composite PCMs. In addition, thermal conductivity of the GP composite was even lower relative to that of pristine PCM due to the presence of macropores in GF. The interconnect structure of porous carbon matrixes with appreciable surface area and intermolecular interactions between porous carbon matrixes and PEG both contribute to a major role for thermal energy storage and the enhanced thermal transfer of composites. CNTs have significantly enhanced the thermal transport of composite PCMs, but the thermal conductivity of composites with CNTs is still not ideal from the perspective of experiment or simulation. The interfacial thermal resistance of CNTs and graphene foam is the bottleneck of heat transfer of composite PCMs. Functional group modification methods or directly growing CNTs on the surface of graphene foam can be employed to enhance the molecular interaction

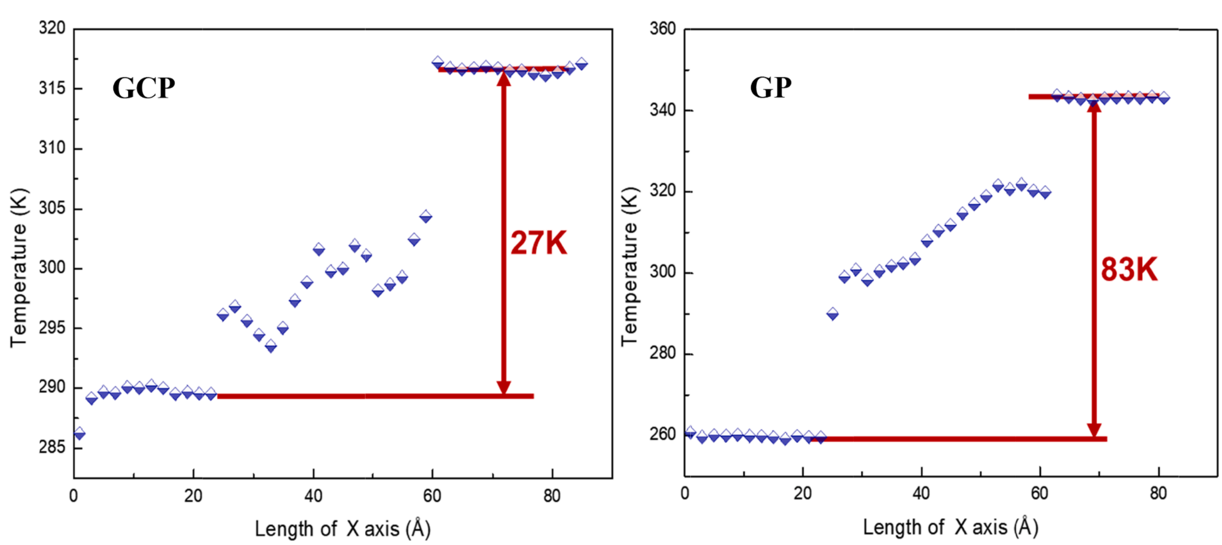


Fig. 12. Temperature distribution of composites before and after adding CNTs (300 K, 25 wt%).

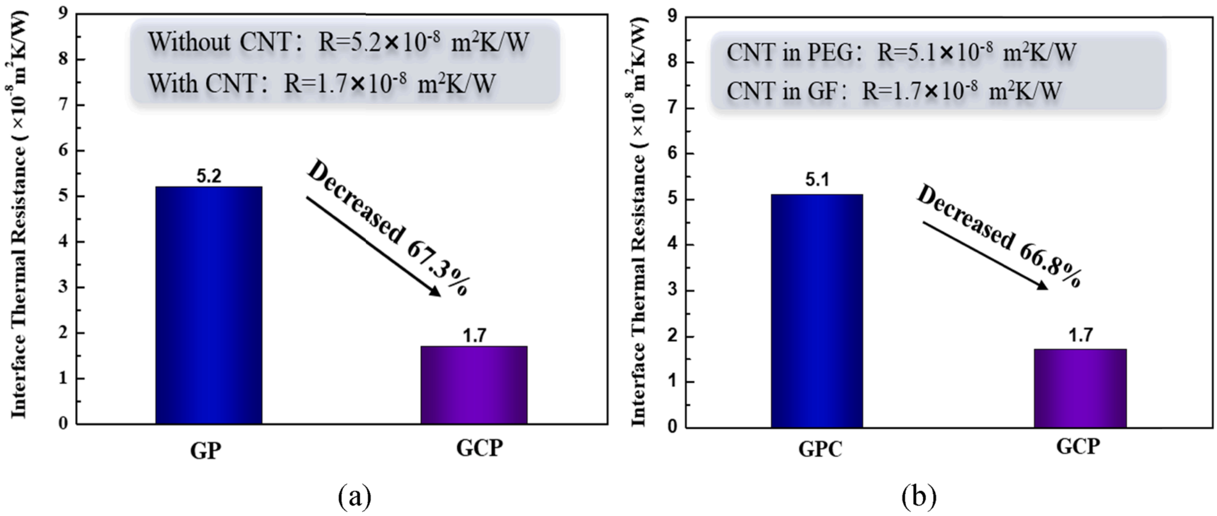


Fig. 13. (a)Interfacial thermal resistance of CNTs before and after composite, (b) Interfacial thermal resistance of GPC and GCP.

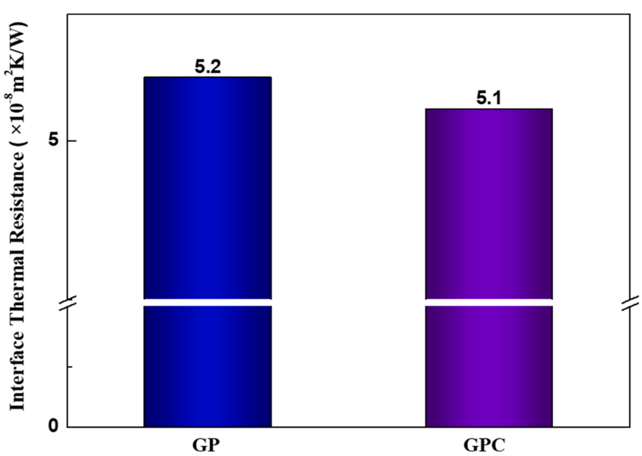


Fig. 14. (a)Interfacial thermal resistance of GP and GPC.

between CNTs and graphene foams, thus improving the thermal conductivity of composite PCMs.

The large thermal energy storage capacity, enhanced thermal conductivity and suitable phase change temperature make these composite PCMs promising candidates for thermal management and storage systems, including solar thermal utilization systems and air-conditioning applications. Thus, the use of these strategies will allow the design of novel composite ssPCMs in response to current global energy consumption demands.

CRediT authorship contribution statement

Zepei Yu: Data curation, Writing – original draft, Visualization, Investigation. **Daili Feng:** Writing – review & editing. **Yanhui Feng:** Supervision, Methodology. **Xinxin Zhang:** Conceptualization, Software.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] Karaman S, Karaipekli A, Sarı A, Biçer A. Polyethylene glycol (PEG)/diatomite composite as a novel form-stable phase change material for thermal energy storage. Sol. Energy Mater. Sol. Cells 2011;95(7):1647–53.
[2] Pielichowska K, Pielichowski K. Phase change materials for thermal energy storage. Prog. Mater Sci. 2014;65:67–123.
[3] Xu Q, Feng J, Zhang S. Influence of end side displacement load on stress and deformation of "L"-type large-diameter buried pipe network. Appl. Therm. Eng. 2017;126:245–54.
[4] Xu Q, Feng J, Zhang S. Combined effects of different temperature and pressure loads on the "L"-type large-diameter buried pipeline. Int. J. Heat Mass Transf. 2017;111:953–61.
[5] Liu H, Fu R, Su X, Wu B, Wang H, Xu Y, et al. MXene confined in shape-stabilized phase change material combining enhanced electromagnetic interference shielding and thermal management capability. Compos. Sci. Technol. 2021;210:108835.
[6] Meng Y, Zhao Y, Zhang Y, Tang B. Induced dipole force driven PEG/PPEGMA form-stable phase change energy storage materials with high latent heat. Chem. Eng. J. 2020;390:124618.
[7] Yu Z, Feng Y, Feng D, Zhang X. Thermal conductance bottleneck of a three dimensional graphene–CNT hybrid structure: a molecular dynamics simulation. PCCP 2020;22(1):337–43.
[8] Aneke M, Wang M. Energy storage technologies and real life applications – A state of the art review. Appl. Energy 2016;179:350–77.
[9] Wu W, Huang X, Li K, Yao R, Chen R, Zou R. A functional form-stable phase change composite with high efficiency electro-to-thermal energy conversion. Appl. Energy 2017;190:474–80.
[10] Liao H, Chen W, Liu Y, Wang Q. A phase change material encapsulated in a mechanically strong graphene aerogel with high thermal conductivity and excellent shape stability. Compos. Sci. Technol. 2020;189:108010.
[11] Atinafu DG, Dong W, Berardi U, Kim S. Phase change materials stabilized by porous metal supramolecular gels: Gelation effect on loading capacity and thermal performance. Chem. Eng. J. 2020;394:124806.
[12] Liu C, Yu W, Chen C, Xie H, Cao B. Remarkably reduced thermal contact resistance of graphene/olefin block copolymer/paraffin form stable phase change thermal interface material. Int. J. Heat Mass Transf. 2020;163:120393.
[13] Huang X, Liu Z, Xia W, Zou R, Han RPS. Alkylated phase change composites for thermal energy storage based on surface-modified silica aerogels. J. Mater. Chem. A 2015;3(5):1935–40.
[14] Xue F, Lu Yu, Qi X-D, Yang J-H, Wang Y. Melamine foam-templated graphene nanoplatelet framework toward phase change materials with multiple energy conversion abilities. Chem. Eng. J. 2019;365:20–9.
[15] Feng D, Feng Y, Li P, Zang Y, Wang C, Zhang X. Modified mesoporous silica filled with PEG as a shape-stabilized phase change materials for improved thermal energy storage performance. Microporous Mesoporous Mater. 2020;292:109756.
[16] Feng D, Feng Y, Qiu L, Li P, Zang Y, Zou H, et al. Review on nanoporous composite phase change materials: Fabrication, characterization, enhancement and molecular simulation. Renew. Sustain. Energy Rev. 2019;109:578–605.
[17] Zalba B, MarÍN JM, Cabeza LF, Mehling H. Review on thermal energy storage with phase change: materials, heat transfer analysis and applications. Applied Thermal Engineering 2003;23:251–83.
[18] Yuan K, Zhou Y, Sun W, Fang X, Zhang Z. A polymer-coated calcium chloride hexahydrate/expanded graphite composite phase change material with enhanced thermal reliability and good applicability. Compos. Sci. Technol. 2018;156:78–86.

[19] Chen L, Zou R, Xia W, Liu Z, Shang Y, Zhu J, et al. Electro- and photodriven phase change composites based on wax-infiltrated carbon nanotube sponges. ACS Nano 2012;6(12):10884–92.
[20] Tang F, Su D, Tang Y, Fang G. Synthesis and thermal properties of fatty acid eutectics and diatomite composites as shape-stabilized phase change materials with enhanced thermal conductivity. Sol. Energy Mater. Sol. Cells 2015;141:218–24.
[21] Alva G, Huang X, Liu L, Fang G. Synthesis and characterization of microencapsulated myristic acid–palmitic acid eutectic mixture as phase change material for thermal energy storage. Appl. Energy 2017;203:677–85.
[22] Dong K, Sheng N, Zou D, Wang C, Nomura T. A high-thermal-conductivity, high-durability phase-change composite using a carbon fibre sheet as a supporting matrix. Appl. Energy 2020;264:114685.
[23] Shah KW. A Review on Enhancement of Phase Change Materials - A Nanomaterials Perspective. Energy Build. 2018;175:57–68.
[24] Wang W, Yang X, Fang Y, Ding J, Yan J. Preparation and thermal properties of polyethylene glycol/expanded graphite blends for energy storage. Appl. Energy 2009;86(9):1479–83.
[25] Chen Y, Ding H, Wang B, Shi Q, Gao J, Cui Z, et al. Dopamine functionalization for improving crystallization behaviour of polyethylene glycol in shape-stable phase change material with silica fume as the matrix. J. Cleaner Prod. 2019;208:951–9.
[26] Mishra AK, Lahiri BB, Philip J. Carbon black nano particle loaded lauric acid-based form-stable phase change material with enhanced thermal conductivity and photo-thermal conversion for thermal energy storage. Energy 2020;191:116572.
[27] Kumar MA, Lahiri BB, John P. Thermal conductivity enhancement in organic phase change material (phenol-water system) upon addition of Al2O3, SiO2 and TiO2 nano-inclusions. J. Mol. Liq. 2018;269:47–63.
[28] Zhang Q, Luo Z, Guo Q, Wu G. Preparation and thermal properties of short carbon fibers/erythritol phase change materials. Energy Convers. Manage. 2017;136:220–8.
[29] Yu S, Jeong S-G, Chung O, Kim S. Bio-based PCM/carbon nanomaterials composites with enhanced thermal conductivity. Sol. Energy Mater. Sol. Cells 2014;120:549–54.
[30] Wei H, Li X. Preparation and characterization of a lauric-myristic-stearic acid/Al2O3-loaded expanded vermiculite composite phase change material with enhanced thermal conductivity. Sol. Energy Mater. Sol. Cells 2017;166:1–8.
[31] Elgafy A, Lafdi K. Effect of carbon nanofiber additives on thermal behavior of phase change materials. Carbon 2005;43(15):3067–74.
[32] Fathinezhad M, AbbasiTarighat M, Dastan D. Chemometrics heavy metal content clusters using electrochemical data of modified carbon paste electrode. Environ. Nanotechnol. Monit. Manage. 2020;14:100307.
[33] Cui Y, Liu C, Hu S, Yu X. The experimental exploration of carbon nanofiber and carbon nanotube additives on thermal behavior of phase change materials. Sol. Energy Mater. Sol. Cells 2011;95(4):1208–12.
[34] Zhong Y, Guo Q, Li S, Shi J, Liu L. Heat transfer enhancement of paraffin wax using graphite foam for thermal energy storage. Sol. Energy Mater. Sol. Cells 2010;94(6):1011–4.
[35] Mills A, Farid M, Selman JR, Al-Hallaj S. Thermal conductivity enhancement of phase change materials using a graphite matrix. Appl. Therm. Eng. 2006;26(14-15):1652–61.
[36] Zhong Y, Zhou M, Huang F, Lin T, Wan D. Effect of graphene aerogel on thermal behavior of phase change materials for thermal management. Sol. Energy Mater. Sol. Cells 2013;113:195–200.
[37] Rehman T-u, Ali HM. Experimental study on the thermal behavior of RT-35HC paraffin within copper and Iron-Nickel open cell foams: Energy storage for thermal management of electronics. Int. J. Heat Mass Transf. 2020;146:118852. https://doi.org/10.1016/j.ijheatmasstransfer.2019.118852.
[38] Thapa S, Chukwu S, Khaliq A, Weiss L. Fabrication and analysis of small-scale thermal energy storage with conductivity enhancement. Energy Convers. Manage. 2014;79:161–70.
[39] Hong S-T, Herling D R. Effects of Surface Area Density of Aluminum Foams on Thermal Conductivity of Aluminum Foam-Phase Change Material Composites. Adv. Eng. Mater. 2007;9(7):554–7.
[40] Li WH, Lai-Iskandar S, Tan D, Simonini L, Dudon J-P, Leong FN, et al. Thermal Conductivity Enhancement and Shape Stabilization of Phase-Change Materials Using Three-Dimensional Graphene and Graphene Powder. Energy Fuels 2020;34(2):2435–44.
[41] Huang X, Alva G, Liu L, Fang G. Preparation, characterization and thermal properties of fatty acid eutectics/bentonite/expanded graphite composites as novel form–stable thermal energy storage materials. Sol. Energy Mater. Sol. Cells 2017;166:157–66.
[42] Feng D, Feng Y, Zang Y, Li P, Zhang X. Phase change in modified metal organic frameworks MIL-101(Cr): Mechanism on highly improved energy storage performance. Microporous Mesoporous Mater. 2019;280:124–32.
[43] Yu Z, Feng Y, Feng D, Zhang X. Thermal properties of three-dimensional hierarchical porous graphene foam-carbon nanotube hybrid structure composites with phase change materials. Microporous Mesoporous Mater. 2020;312:110781.
[44] Nomura T, Tabuchi K, Zhu C, Sheng N, Wang S, Akiyama T. High thermal conductivity phase change composite with percolating carbon fiber network. Appl. Energy 2015;154:678–85.
[45] Zhao YJ, Wang RZ, Wang LW, Yu N. Development of highly conductive KNO3/NaNO3 composite for TES [thermal energy storage]. Energy 2014;70:272–7.
[46] Dastan D. Effect of preparation methods on the properties of titania nanoparticles: solvothermal versus sol–gel. Appl. Phys. A 2017;123:699.
[47] Feng DL, Zang YY, Pei LI, Feng YH, Zhang XX. Polyethylene glycol phase change material embedded in a hierarchical porous carbon with superior thermal storage capacity and excellent stability. Compos. Sci. Technol. 2021;210:108832.
[48] Tersoff J. Modeling solid-state chemistry: Interatomic potentials for multicomponent systems. Physical Review B 1989;39:5566–8.
[49] Tang J, Yang M, Dong W, Yang M, Zhang H, Fan S, et al. Highly porous carbons derived from MOFs for shape-stabilized phase change materials with high storage capacity and thermal conductivity. RSC Adv. 2016;6(46):40106–14.
[50] Tan G-L, Tang D, Dastan D, Jafari A, Shi Z, Chu Q-Q, et al. Structures, morphological control, and antibacterial performance of tungsten oxide thin films. Ceram. Int. 2021;47(12):17153–60.
[51] Tan G-L, Tang D, Dastan D, Jafari A, Silva JPB, Yin X-T. Effect of heat treatment on electrical and surface properties of tungsten oxide thin films grown by HFCVD technique. Mater. Sci. Semicond. Process. 2021;122:105506.
[52] Gao X, Xie B, Su Y, Fu D, Wang D. Nanoparticle Enlarged Interfacial Effect on Phase Transition of 1-Octadecanol/Silica Composites. J. Phys. Chem. B 2015;119(5):2074–80.
[54] Antoniammal P, Arivuoli D. Size and Shape Dependence on Melting Temperature of Gallium Nitride Nanoparticles. J. Nanomaterials 2012;2012:1–11.
[55] Luan Y, Yang M, Ma Q, Qi Y, Gao H, Wu Z, et al. Introduction of an organic acid phase changing material into metal–organic frameworks and the study of its thermal properties. J. Mater. Chem. A 2016;4(20):7641–9.
[56] Li A, Wang J, Dong C, Dong W, Atinafu DG, Chen X, et al. Core-sheath structural carbon materials for integrated enhancement of thermal conductivity and capacity. Appl. Energy 2018;217:369–76.
[57] Nika DL, Askerov AS, Balandin AA. Anomalous size dependence of the thermal conductivity of graphene ribbons. Nano Lett. 2012;12(6):3238–44.
[58] Shan K, Yi Z-Z, Yin X-T, Dastan D, Dadkhah S, Coates BT, et al. Mixed conductivities of A-site deficient Y, Cr-doubly doped SrTiO3 as novel dense diffusion barrier and temperature-independent limiting current oxygen sensors. Adv. Powder Technol. 2020;31(12):4657–64.
[59] Shan K, Zhai F, Yi Z-Z, Yin X-T, Dastan D, Tajabadi F, et al. Mixed conductivity and the conduction mechanism of the orthorhombic CaZrO3 based materials. Surf. Interfaces 2021;23:100905.
[60] Wei S, Shi Z, Wei W, Wang H, Dastan D, Huang M, et al. Facile preparation of ultralight porous carbon hollow nanoboxes for electromagnetic wave absorption. Ceram. Int. 2021;47(19):28014–20.